

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 3: Due by midnight on Saturday February 10, 2024.

Submit to WyoCourses a zipped folder containing a Jupyter notebook (both ipynb and HTML files).

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [40 points] Consider the following system of equations called the Fitzhugh-Nagumo model for a neuron

$$\begin{aligned}\frac{dv}{dt} &= f_1(t, v, w) = v - v^3 - w + I_{ext} \\ \frac{dw}{dt} &= f_2(t, v, w) = (v + a - bw)/\tau.\end{aligned}$$

Here, $v(t)$ and $w(t)$ represent a neuron membrane potential (i.e., voltage) and a recovery variable. I_{ext} is an external stimulus (i.e., current of electricity), and we will fix the constants $a = 0.3$, $b = 1.4$ and $\tau = 20$.

- (A). [10 points] Fix $I_{ext} = 0$ and initial condition $v(0) = w(0) = 0$, and use the forward Euler method to numerically solve the IVP for $t \in [0, 500]$ with $N = 2000$ time steps. Plot $v(t)$ and $w(t)$ versus t . Then, repeat the simulation but change the stimulus to $I_{ext} = 0.2$. Describe how your results change.
- (B). [10 points] Repeat Problem 1A but instead use the midpoint method, comparing your approximate solution to that for the FE method.
- (C). [10 points] Repeat Problem 2B but use the 4th order Runge-Kutta method (RK4) instead of the midpoint method. Again, compare your results to those from the FE method.
- (D). [10 points] One can approximate the simulation errors for the FE and midpoint methods by comparing them to results for the RK4 method (which is much more accurate.) For both $I_{ext} = 0$ and $I_{ext} = 0.2$, plot the approximated global errors

$$E_i = \|\mathbf{y}_i^{(FE)} - \mathbf{y}_i^{(RK)}\|_2 = \sqrt{(v_i^{(FE)} - v_i^{(RK)})^2 + (w_i^{(FE)} - w_i^{(RK)})^2},$$

where we define the vector $\mathbf{y}_i = [v_i, w_i]$ and $\|\cdot\|_2$ is the vector 2-norm. Instead of using `plot()`, I suggest using the Python function `semilogy()` so that the vertical axis has a log scale.

2. [40 points] Consider the initial value problem (IVP)

$$\begin{aligned}\frac{dy}{dt} &= f(t, y(t)) = -20[y(t) - \cos(t)] - \sin(t) \\ y(0) &= 1.5.\end{aligned}$$

- (A). [20 points] Numerically integrate the IVP for $t \in [0, 1]$ using the midpoint method for $N = 32$ steps. Make a plot that compares your predictions $y_i \approx y(t_i)$ for $t_i = i/n$ and $i = 0, 1, \dots, n$ to the true function: $y(t) = 0.5e^{-20t} + \cos(t)$. Make a second plot showing the global error $E_i = |y(t_i) - y_i|$ at each time t_i .
- (B). [10 points] Repeat Problem 1A for both the forward Euler and Backward Euler methods. Depict your results using two plots. The first should show all 3 approximations (i.e., using the FE, BE and midpoint methods) as different colored dots, and you should plot the true solution for comparison. The second plot should show the global errors versus t_i for each of the three methods.
- (C). [10 points] In two new figures, repeat Problems 2B but use both $N = 16$ and $N = 8$. Describe what happens as N decreases. Which method performs best for $N = 32$ and why? Which method seems to perform best and why? *Hint: The IVP is called a stiff problem since $y(t)$ has both fast and slow dynamics.*

3. [20 points]

Consider the linear differential equation $y' = ay + b$ with $a < 0$.

(A). [5 points] Prove that $y = y^* = -b/a$ is an equilibrium.

(B). [10 points] Write down the Backward Euler (BE) method for this IVP and rearrange the equation into the form $y_{i+1} = g(y_i)$

(C). [5 points] Interpret this equation as a fixed point iteration and prove $y_i \rightarrow y^*$ as i to infinity.